Exact Information Geometry and New Frontiers

for the Fabius Function, Rvachev's Up-Function, Thue–Morse Sinc Products, and Geometric q-Convolutions

Research report prepared for Vladimir Reshetnikov

Repository-aware analysis, exact proofs, and reproducible experiments

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Abstract

The Fabius–Rvachev corpus already connects the Thue–Morse sequence, exact dyadic arithmetic, q-binomial interpolation, infinite sinc products, Bell–Bernoulli moment calculus, Legendre expansions, inverse functions, and lower-Lambert endpoint asymptotics. This report develops a different, previously unexploited organizing layer: the exact information carried by each geometric convolution scale.

For

$$X_q = (1-q)\sum_{j\geq 0} q^j U_j, \qquad U_j \overset{\text{iid}}{\sim} \mathrm{Unif}[-1,1], \qquad 0<q<1,$$

let $S_{q,m}$ be the first m summands. The self-similar tail identity $X_q = S_{q,m} + q^m X_q'$ implies the exact non-Gaussian channel law

$$I(X_q; S_{q,m}) = m\log(1/q), \qquad \mathrm{mmse}(X_q \mid S_{q,m}) = q^{2m}\,\mathrm{Var}(X_q),$$

and, strikingly,

$$I(X_q; S_{q,m}) = -\frac{1}{2}\log\bigl(1-\mathrm{Corr}(X_q, S_{q,m})^2\bigr),$$

the familiar Gaussian correlation formula holding here for a compactly supported, nowhere-analytic density. Every new geometric digit contributes exactly $\log(1/q)$ nats conditionally. The scalar prefix is a sufficient statistic for the entire digit block, and its posterior is an exact translated, rescaled copy of the original Fabius–Rvachev law. This yields closed posterior cumulants, Bell moments, quantiles, Rényi entropies, Fisher information, and transported Legendre expansions.

At $q=1/2$, the finite prefix density is a Thue–Morse signed spline. Substitution into the posterior formula gives an exact Kullback–Leibler integral equaling $m\log 2$, thereby joining a finite Thue–Morse product to the infinite Rvachev sinc product in one Bayesian identity. The centered information density converges to a difference of two independent Rvachev surprisals; its limiting variance is twice the varentropy and its cumulant generating function is a symmetric pair of Rényi integrals.

A second main theorem identifies the first entropy correction of the finite sinc prefixes:

$$h(X_q) - h(S_{q,m}) = \frac{\mathrm{Var}(X_q)\,\mathcal{J}(f_q)}{2} q^{2m} + o(q^{2m}),$$

where $\mathcal{J}$ is Fisher information. In the dyadic case a new bridge gives

$$\mathcal{J}(\mathrm{up}) = \int_0^1 \frac{F'(x)^2}{F(x)(1-F(x))}\,\mathrm{d}x = \int_0^1 \frac{\mathrm{d}y}{y(1-y)(F^{-1})'(y)}.$$

Combining the inverse-Fabius elasticity with the lower-Lambert scale produces a new endpoint entropy law. Finally, exact Bernoulli cumulants yield a formal entropic Edgeworth expansion as $q\uparrow 1$; the report isolates the remaining uniform-analysis step as a precise conjecture and tests it numerically. All claims are status-labeled, and the archive contains the complete LaTeX source, compiled PDF, commented Python code, exact symbolic checks, CSV data, and vector figures.

Contents

1 Scope, corpus audit, and contribution boundary

1.1 What was audited

The source audit covered the recursive LaTeX corpus under `Analysis/FabiusFunction/docs` in the current `ProveIt` tree, including the primary exposition, glossary, Lean walkthrough, active semi-formalized and non-formalized frontier volumes, archived studies, representation drafts, inverse-and-sampling drafts, q-series drafts, and the recent parallel reports mirrored in the working library. The audit was claim-oriented rather than merely title-oriented: theorem statements, conclusion sections, imported-interface lists, and exact or close-variant phrase searches were checked for the proposed notions "mutual information", "MMSE", "rate distortion", "information density", "varentropy", "finite-prefix entropy", and "Fisher information".

The following are treated strictly as imported inputs and are not relabeled as new:

- the random-uniform series for the Fabius/Rvachev law, its refinement and differential-dilation equations, and its infinite sinc product;
- exact dyadic values, Thue–Morse derivative and truncated-power formulas, rational moments, and the Bell–Bernoulli cumulant calculus;
- the geometric-q family, Gaussian-binomial interpolation, reciprocal-q deconvolution, negative-q affine dualities, and finite convolution models;
- inverse-Fabius algebraic germs, inverse derivative recurrences, and the phase-aware all-orders lower-Lambert endpoint calculus;
- Legendre, Jacobi, Lagrange, sampling, multiresolution, Fredholm, resolvent, Appell, Stein, and Koopman representations already developed in the corpus;
- the existing global entropy identity $\int_0^1 \log(F^{-1})'(y)\,\mathrm{d}y = h(F')$ and entropy laws for derivative-energy measures in the non-dyadic atomic-function hierarchy.

The phrase "new" below therefore means *not located in the audited repository corpus under the displayed normalization.* It is not an unconditional claim of worldwide publication priority. Several abstract ingredients, such as Shannon's lower bound, entropy power, high-resolution rate distortion, Fisher information, and entropic Edgeworth expansions, have substantial external literatures. The frontier contribution is their exact synthesis with the geometric Fabius–Rvachev convolution and its Thue–Morse, inverse, and Lambert structures.

1.2 Result ledger

Result	Status	Main mechanism
Exact prefix mutual information $I(X_q; S_{q,m}) = m\log(1/q)$	proved	independent self-similar tail
Constant information per geometric digit	proved	chain rule and prefix sufficiency
Posterior location-scale copy, quantiles, cumulants, Bell moments, Rényi and Fisher scaling	proved	conditional density formula
Non-Gaussian Gaussian-correlation identity and exact MMSE	proved	orthogonal projection onto the prefix

Result	Status	Main mechanism
Rate-distortion bracket and asymptotic prefix-code gap	derived	Shannon lower bound and high-resolution theory
Finite Thue–Morse spline Bayesian/KL identity	proved	inclusion-exclusion spline plus Rvachev posterior
Information-spectrum limit and Rényi cumulant generating function	proved	independent-tail coupling
Finite-sinc entropy correction governed by Fisher information	proved	Fourier deconvolution and entropy variation
Fisher bridge through F, F^{-1}, and up	proved	symmetry, differentiation, and quantile transport
Lower-Lambert endpoint tail-entropy law	derived from audited asymptotics	inverse elasticity and Watson integration
Entropic Edgeworth coefficients $3/100$ and $33/500$ near $q=1$	formal plus numerical	Bernoulli cumulants and Hermite orthogonality
Full entropic $q \uparrow 1$ expansion and monotonic deficit	conjectural	uniform Fourier/entropy control

Central new principle. The geometric convolution is not only a product representation of a density. It is an exact equal-information filtration: observing one additional uniform scale always reveals precisely $\log(1/q)$ nats about the final random variable, independently of the depth and despite the strongly non-Gaussian posterior.

2 The geometric-uniform family and the dyadic dictionary

2.1 Probability law, fixed point, and Fourier product

Let $U_0, U_1, \ldots$ be independent $\mathrm{Unif}[-1,1]$ random variables and set

$$X_q = (1-q)\sum_{j=0}^{\infty} q^j U_j, \qquad 0<q<1. \tag{1}$$

The series converges absolutely and $X_q \in [-1,1]$. Write f_q for its density, F_q for its distribution function, and

$$\phi_q(t) = \mathbb{E}\mathrm{e}^{\mathrm{i}tX_q} = \prod_{j=0}^{\infty} \operatorname{sinc}\big((1-q)q^j t\big), \qquad \operatorname{sinc} z = \frac{\sin z}{z}. \tag{2}$$

The first digit gives the distributional fixed point

$$X_q \overset{d}{=} qX_q' + (1-q)U_0, \tag{3}$$

where X_q' is an independent copy. Consequently

$$f_q(x) = \frac{F_q((x+1-q)/q) - F_q((x-1+q)/q)}{2(1-q)}. \tag{4}$$

The product in (2) decays faster than every power on the real line; thus $f_q \in C_c^{\infty}(\mathbb{R})$ and is flat at the support endpoints. These regularity facts are repository inputs, but they are also immediate from the standard “number of active sinc factors” estimate.

2.2 Variance, Bernoulli cumulants, and Bell moments

Since $\mathrm{Var}(U_j) = 1/3$,

$$\sigma_q^2 := \mathrm{Var}(X_q) = \frac{(1-q)^2}{3} \sum_{j\geq 0} q^{2j} = \frac{1-q}{3(1+q)}. \tag{5}$$

The logarithm of $\sinh z/z$ gives the even cumulants

$$\kappa_{2r}(X_q) = \frac{2^{2r} B_{2r}}{2r} \frac{(1-q)^{2r}}{1-q^{2r}}, \qquad \kappa_{2r+1}(X_q) = 0. \tag{6}$$

Here B_n are the Bernoulli numbers with $B_2 = 1/6$. Every ordinary moment is therefore a complete Bell polynomial:

$$\mu_n(q) := \mathbb{E}X_q^n = \mathcal{B}_n(0, \kappa_2, 0, \kappa_4, \ldots, \kappa_n). \tag{7}$$

This imported Bell–Bernoulli interface will make the posterior moment formulas below completely explicit.

2.3 The Fabius–Rvachev specialization

At $q = 1/2$, write $X = X_{1/2}$. In the probability normalization used here,

$$\mathbb{P}(X \in \mathrm{d}x) = \mathrm{up}(x)\,\mathrm{d}x, \qquad -1 \leq x \leq 1, \tag{8}$$

and the affine variable $Y = (X+1)/2$ has the Fabius distribution:

$$\mathbb{P}(Y \leq y) = F(y), \qquad F'(y) = 2\,\mathrm{up}(2y-1), \qquad 0 < y < 1. \tag{9}$$

Equivalently,

$$\mathrm{up}(x) = F(1 - |x|), \qquad |x| \leq 1. \tag{10}$$

Let

$$G = F^{-1} : (0,1) \longrightarrow (0,1) \tag{11}$$

be the inverse Fabius function. The variance is

$$\mathrm{Var}(X) = \frac{1}{9}. \tag{12}$$

The report will move repeatedly among the centered density up, the bounded CDF F, and the quantile G; (9) fixes every factor of two.

3 An exact equal-information filtration

3.1 Prefix and tail decomposition

For $m \geq 1$, define the finite geometric prefix

$$S_{q,m} = (1-q) \sum_{j=0}^{m-1} q^j U_j. \tag{13}$$

Let $X_q^{(m)}$ be an independent copy of X_q, formed from the tail digits. Then

$$X_q = S_{q,m} + q^m X_q^{(m)}, \qquad S_{q,m} \perp X_q^{(m)}. \tag{14}$$

This elementary identity is the engine behind all exact information formulas.

Theorem 3.1 (Exact prefix information). *For every $0 < q < 1$ and integer $m \geq 1$,*

$$\boxed{I(X_q; S_{q,m}) = m \log(1/q).} \tag{15}$$

Moreover the full digit block and its scalar sum carry exactly the same information:

$$I(X_q; U_0, \ldots, U_{m-1}) = I(X_q; S_{q,m}). \tag{16}$$

Proof. Conditioned on $S_{q,m} = s$, equation (14) says that X_q is distributed as $s + q^m X_q^{(m)}$. Translation does not alter differential entropy and multiplication by q^m adds $m \log q$. Hence

$$h(X_q \mid S_{q,m}) = h(X_q) + m \log q,$$

which gives (15). Conditioned on the entire digit block, $S_{q,m}$ is known and the same translated tail remains. Therefore $h(X_q \mid U_0, \ldots, U_{m-1}) = h(X_q \mid S_{q,m})$, proving (16). □

Corollary 3.2 (Constant information per scale). *Every additional geometric digit contributes exactly one constant increment:*

$$I\big(X_q; U_{m-1} \mid U_0, \ldots, U_{m-2}\big) = \log(1/q). \tag{17}$$

At $q = 1/2$, the increment is exactly one bit and

$$I(X; S_{1/2,m}) = m \log 2 = m \text{ bits}. \tag{18}$$

Proof. Apply the chain rule for mutual information and subtract the depth $m-1$ identity from the depth m identity in Theorem 3.1. □

Remark 3.3 (Why the information can exceed the entropy of the source). Differential entropy is not a bound on mutual information. As $m \to \infty$, the posterior width tends to zero, so the continuous observation becomes arbitrarily precise and $I(X_q; S_{q,m}) \to \infty$. The linear law is the exact continuous analogue of gaining a fixed number of radix digits per scale.

3.2 The posterior is a self-similar copy

Theorem 3.4 (Posterior location-scale calculus). *For every prefix value s in the support of $S_{q,m}$,*

$$p_{X_q \mid S_{q,m}=s}(x) = q^{-m} f_q\left(\frac{x-s}{q^m}\right). \tag{19}$$

Consequently, for $0 < p < 1$,

$$Q_{X_q \mid s}(p) = s + q^m Q_q(p), \tag{20}$$

where $Q_q = F_q^{-1}$, and

$$\mathbb{E}\big[\mathrm{e}^{\mathrm{i}tX_q} \mid S_{q,m} = s\big] = \mathrm{e}^{\mathrm{i}ts} \phi_q(q^m t). \tag{21}$$

The scalar $S_{q,m}$ is a sufficient statistic for the entire block $(U_0, \ldots, U_{m-1})$ relative to X_q.

Proof. All statements follow by conditioning in (14). The factorization criterion for sufficiency is literal: the conditional law of X_q given the digit block depends on that block only through $S_{q,m}$. □

Corollary 3.5 (Posterior cumulants and Bell moments)**.** *For $r \geq 2$,*

$$\kappa_r(X_q \mid S_{q,m} = s) = q^{mr}\kappa_r(X_q), \tag{22}$$

while the conditional mean is s. In particular,

$$\mathbb{E}\big[(X_q - s)^n \mid S_{q,m} = s\big] = q^{mn}\mathcal{B}_n(0, \kappa_2, 0, \kappa_4, \ldots, \kappa_n). \tag{23}$$

Thus every posterior central moment is a Bernoulli–Bell expression multiplied by one geometric power.

Corollary 3.6 (Posterior entropy, Rényi entropy, Fisher information, and varentropy)**.** *Let h_α denote differential Rényi entropy of order $\alpha > 0$, $\alpha \neq 1$, and let*

$$\mathcal{J}(f_q) = \int_{-1}^{1} \frac{f_q'(x)^2}{f_q(x)}\,\mathrm{d}x, \qquad V(f_q) = \mathrm{Var}[-\log f_q(X_q)].$$

Then

$$h(X_q \mid S_{q,m} = s) = h(X_q) + m\log q, \tag{24}$$

$$h_\alpha(X_q \mid S_{q,m} = s) = h_\alpha(X_q) + m\log q, \tag{25}$$

$$\mathcal{J}(X_q \mid S_{q,m} = s) = q^{-2m}\mathcal{J}(f_q), \tag{26}$$

$$V(X_q \mid S_{q,m} = s) = V(f_q). \tag{27}$$

The posterior contracts geometrically in space, loses $\log(1/q)$ *nats of entropy per scale, gains q^{-2} in Fisher information per scale, and preserves varentropy exactly.*

Proof. Translation has no effect on these functionals. Under $x \mapsto ax$, every Rényi entropy gains $\log|a|$, Fisher information is multiplied by a^{-2}, and the surprisal changes only by the additive constant $\log|a|$; hence its variance is invariant. □

3.3 Transport of Legendre and Appell representations

The audited repository contains Legendre expansions of up and inverse-moment Appell polynomial systems for f_q. The posterior theorem transports either representation without recomputing a coefficient.

Proposition 3.7 (Posterior Legendre transport)**.** *Suppose*

$$f_q(z) = \sum_{n\geq 0} c_n(q)P_n(z) \tag{28}$$

converges in any mode valid for the repository's expansion. Then

$$p_{X_q|s}(x) = q^{-m}\sum_{n\geq 0} c_n(q)P_n\left(\frac{x-s}{q^m}\right) \tag{29}$$

for $|x - s| \leq q^m$. In particular, all odd coefficients vanish by symmetry, and at $q = 1/2$ the existing even Legendre expansion of up *becomes an exact local posterior expansion at every prefix value.*

Proof. Substitute (28) into (19). No mixing of Legendre modes occurs in the normalized posterior coordinate. □

Remark 3.8 (A Bayesian meaning of the Appell deconvolution). If $A_n^{(q)}$ denotes the repository's reciprocal-moment Appell family for X_q, then the random residual $(X_q - S_{q,m})/q^m$ has exactly the f_q law. Therefore every mean-zero Appell identity for X_q holds conditionally after this normalization. The posterior is not merely similar in shape; it carries the same complete polynomial deconvolution algebra at every depth.

4 Estimation, correlation, and rate distortion

4.1 Exact MMSE and a Gaussian-form identity without Gaussianity

Theorem 4.1 (Exact posterior mean and MMSE)**.** *For every $m \geq 1$,*

$$\mathbb{E}[X_q \mid S_{q,m}] = S_{q,m}, \qquad \mathrm{Var}(X_q \mid S_{q,m}) = q^{2m}\sigma_q^2, \tag{30}$$

and hence

$$\boxed{\mathrm{mmse}(X_q \mid S_{q,m}) = q^{2m}\sigma_q^2.} \tag{31}$$

Moreover

$$\mathrm{Corr}(X_q, S_{q,m})^2 = 1 - q^{2m}. \tag{32}$$

Proof. The tail in (14) is independent, centered, and has variance $q^{2m}\sigma_q^2$. Thus the prefix is the conditional mean and is the orthogonal L^2 projection. Also

$$\mathrm{Var}(S_{q,m}) = \sigma_q^2 - q^{2m}\sigma_q^2, \qquad \mathrm{Cov}(X_q, S_{q,m}) = \mathrm{Var}(S_{q,m}),$$

which gives (32). □

Corollary 4.2 (Gaussian-correlation formula for a compact non-Gaussian channel)**.** *The exact information law can be rewritten as*

$$\boxed{I(X_q; S_{q,m}) = -\frac{1}{2}\log(1 - \mathrm{Corr}(X_q, S_{q,m})^2).} \tag{33}$$

Proof. By (32), $1 - \mathrm{Corr}^2 = q^{2m}$; insert this in (15). □

Remark 4.3. For a general pair of random variables, correlation does not determine mutual information. Formula (33) is usually associated with a jointly Gaussian pair. Here both variables have compact support, and the density at $q = 1/2$ is nowhere analytic. The equality is forced instead by exact self-similar posterior scaling.

In the dyadic case,

$$\mathrm{mmse}(X \mid S_{1/2,m}) = \frac{4^{-m}}{9}. \tag{34}$$

Thus the m bits in (18) reduce mean-square uncertainty by the exact factor 4^{-m}.

4.2 A self-similar test channel for quadratic rate distortion

Let $R_{X_q}(D)$ be the rate-distortion function of X_q under squared error. Use the deterministic reproduction $\widehat{X}_m = S_{q,m}$. Its rate and distortion are

$$R_m := I(X_q; \widehat{X}_m) = m\log(1/q), \qquad D_m := \mathbb{E}(X_q - \widehat{X}_m)^2 = q^{2m}\sigma_q^2. \tag{35}$$

Therefore

$$D_m = \sigma_q^2 \mathrm{e}^{-2R_m}. \tag{36}$$

Define the Gaussian entropy deficit

$$\delta_q := \frac{1}{2}\log(2\pi\mathrm{e}\sigma_q^2) - h(X_q) = D_{\mathrm{KL}}\left(f_q \,\middle\|\, \frac{1}{\sqrt{2\pi\sigma_q^2}}\exp\left[-\frac{x^2}{2\sigma_q^2}\right]\right). \tag{37}$$

It is nonnegative and vanishes only for a Gaussian law.

Theorem 4.4 (Rate-distortion bracket at geometric scales). *For every $m \geq 1$,*

$$R_m - \delta_q \leq R_{X_q}(D_m) \leq R_m. \tag{38}$$

Moreover, as $m \to \infty$,

$$R_{X_q}(D_m) = R_m - \delta_q + o(1). \tag{39}$$

Thus the asymptotic rate penalty of the exact geometric-prefix reproduction is precisely the non-Gaussian entropy deficit δ_q.

Proof. The upper bound is achieved by $\widehat{X}_m = S_{q,m}$. Shannon's lower bound for squared error gives

$$R_{X_q}(D) \geq h(X_q) - \frac{1}{2}\log(2\pi \mathrm{e} D).$$

At $D = D_m$, this equals $R_m - \delta_q$, proving the bracket. The compactly supported smooth density f_q satisfies the standard regularity assumptions for the high-resolution formula

$$R_{X_q}(D) = h(X_q) - \frac{1}{2}\log(2\pi \mathrm{e} D) + o(1) \qquad (D \downarrow 0).$$

Substitution of D_m proves (39). □

Corollary 4.5 (A universal entropy-power bound). *For every $0 < q < 1$,*

$$0 < \delta_q \leq \frac{1}{2}\log\frac{\pi \mathrm{e}}{6}. \tag{40}$$

The upper endpoint is the entropy deficit of a standardized uniform law and is approached as $q \downarrow 0$.

Proof. Apply the entropy-power inequality to the independent summands $(1-q)q^j U_j$. Since $h(U_j) = \log 2$,

$$N(X_q) := \frac{\mathrm{e}^{2h(X_q)}}{2\pi \mathrm{e}} \geq \sum_{j\geq 0}(1-q)^2 q^{2j}\frac{2}{\pi \mathrm{e}} = \frac{2(1-q)}{\pi \mathrm{e}(1+q)}.$$

Compare this with (5). Rearrangement gives (40). Strict positivity follows because f_q has compact support and is not Gaussian. □

Remark 4.6 (Operational meaning). At high resolution the optimal distortion at rate R_m is asymptotic to $D_m \mathrm{e}^{-2\delta_q}$. Hence the exact geometric prefix has multiplicative high-resolution loss $\mathrm{e}^{2\delta_q}$. Numerically, for $q = 1/2$ this factor is approximately 1.0476742: the dyadic prefix is within about 4.77% of the high-resolution optimum while retaining an exact algebraic interpretation.

5 Finite sinc products and a Thue–Morse Bayesian identity

5.1 The finite q-product channel

Let $b_{q,m}$ be the density of $S_{q,m}$. Its characteristic function is the finite sinc product

$$\widehat{b}_{q,m}(t) = \prod_{j=0}^{m-1} \operatorname{sinc}\big((1-q)q^j t\big) = \frac{\phi_q(t)}{\phi_q(q^m t)}. \tag{41}$$

The quotient is interpreted by cancellation at common zeros; the finite product is the primary definition. Taking logarithms or using independent cumulants gives an exact truncation rule.

Proposition 5.1 (Finite-prefix cumulants)**.** *For every $r \geq 1$,*

$$\kappa_r(S_{q,m}) = (1 - q^{rm})\kappa_r(X_q). \tag{42}$$

In particular,

$$\mathrm{Var}(S_{q,m}) = (1 - q^{2m})\sigma_q^2, \tag{43}$$

and for even order $2r$,

$$\kappa_{2r}(S_{q,m}) = \frac{2^{2r}B_{2r}}{2r}\frac{(1-q)^{2r}(1-q^{2rm})}{1-q^{2r}}. \tag{44}$$

Proof. Cumulants add under independent convolution and scale homogeneously. Sum the finite geometric progression in the rth powers of the coefficients. □

The factor $(1 - q^{rm})/(1 - q^r)$ is a q^r-integer. Thus the finite Bayesian channel carries the same geometric factors that appear elsewhere in the corpus as q-integers, Gaussian-binomial coefficients, and finite q-Pochhammer products. Here they govern cumulant truncation rather than interpolation.

5.2 The dyadic Thue–Morse spline

Set $q = 1/2$ and abbreviate

$$S_m := S_{1/2,m} = \sum_{k=1}^{m} 2^{-k}U_{k-1}, \qquad A_m := \sum_{k=1}^{m} 2^{-k} = 1 - 2^{-m}. \tag{45}$$

For $0 \leq j < 2^m$, let

$$\varepsilon_j = (-1)^{s_2(j)}, \tag{46}$$

where $s_2(j)$ is the binary digit sum.

Theorem 5.2 (Exact Thue–Morse prefix density)**.** *The density of S_m is the compact spline*

$$\boxed{b_m(s) = \frac{2^{m(m-1)/2}}{(m-1)!}\sum_{j=0}^{2^m-1}\varepsilon_j\left(s + A_m - \frac{j}{2^{m-1}}\right)_+^{m-1}.} \tag{47}$$

Its Fourier transform is

$$\widehat{b}_m(t) = \prod_{k=1}^{m}\mathrm{sinc}(t/2^k). \tag{48}$$

Proof. The convolution of uniforms on $[-a_k, a_k]$ has the inclusion-exclusion density

$$\frac{1}{(m-1)!\prod_{k=1}^{m}(2a_k)}\sum_{E\subseteq\{1,\ldots,m\}}(-1)^{|E|}\left(s + \sum_k a_k - 2\sum_{k\in E}a_k\right)_+^{m-1}.$$

For $a_k = 2^{-k}$,

$$\prod_{k=1}^{m}(2a_k) = 2^{-m(m-1)/2}.$$

Writing the subset indicator as the binary expansion of j gives $2\sum_{k\in E}2^{-k} = j/2^{m-1}$ and $(-1)^{|E|} = (-1)^{s_2(j)}$. The Fourier product follows directly from convolution. □

Remark 5.3 (Prouhet cancellation is analytically essential)*.* The individual truncated powers in (47) grow polynomially outside the support; only the Thue–Morse cancellation makes their sum compactly supported. This is the finite-channel counterpart of the cancellation phenomena that make direct asymptotics of exact dyadic Fabius formulas delicate.

5.3 Joint density and an exact KL integral

The posterior theorem gives a literal composition of the finite Thue–Morse spline and the infinite sinc-product density.

Theorem 5.4 (Finite/infinite sinc Bayesian bridge)**.** *For the dyadic law $X \sim \mathrm{up}$,*

$$p_{S_m,X}(s,x) = b_m(s)\, 2^m \,\mathrm{up}(2^m(x-s)). \tag{49}$$

Consequently,

$$\boxed{\iint_{\mathbb{R}^2} b_m(s)\, 2^m \,\mathrm{up}(2^m(x-s)) \log \frac{2^m \,\mathrm{up}(2^m(x-s))}{\mathrm{up}(x)}\, \mathrm{d}x\, \mathrm{d}s = m \log 2.} \tag{50}$$

Substituting (47) makes the left side an explicit finite Thue–Morse sum coupled to the infinite Rvachev product.

Proof. Equation (49) is the product of the prefix density and the conditional density from (19). The logarithm in (50) is the likelihood ratio $p_{X|S_m}(x \mid s)/p_X(x)$. Its joint expectation is $I(X;S_m) = m \log 2$ by Theorem 3.1. □

A closed loop among three repository themes. The same depth m has three simultaneous meanings:

1. it is the number of factors in the finite sinc product $\prod_{k=1}^{m} \mathrm{sinc}(t/2^k)$;
2. it is the length of the Thue–Morse inclusion-exclusion block in (47);
3. it is exactly the number of bits of mutual information in (50).

The analytic cancellation, harmonic factorization, and information gain are three views of the same finite convolution.

5.4 Conditional inverse-Fabius quantiles

Let $G = F^{-1}$. The centered Rvachev quantile is

$$Q_{\mathrm{up}}(p) = 2G(p) - 1. \tag{51}$$

Therefore the exact dyadic posterior quantile is

$$Q_{X|S_m=s}(p) = s + 2^{-m}(2G(p) - 1). \tag{52}$$

Every inverse-Fabius endpoint expansion in the repository therefore generates, without further reversion, a full posterior endpoint expansion at every finite Thue–Morse prefix. The lower-Lambert phase is universal across all posterior cells; only the affine center and scale change.

6 Information density, varentropy, and the surprisal limit

6.1 Exact information density

Let $X_q^{(m)}$ be the independent tail copy in (14). The information density of the prefix channel is

$$\imath_m(X_q; S_{q,m}) := \log \frac{p_{X_q|S_{q,m}}(X_q \mid S_{q,m})}{f_q(X_q)}. \tag{53}$$

Theorem 6.1 (Exact information-density decomposition)**.** *Almost surely,*

$$\boxed{\imath_m(X_q; S_{q,m}) = m\log(1/q) + \log f_q(X_q^{(m)}) - \log f_q(X_q).} \tag{54}$$

The second term is exactly centered:

$$\mathbb{E}[\log f_q(X_q^{(m)}) - \log f_q(X_q)] = 0. \tag{55}$$

Proof. At the realized point $X_q = S_{q,m} + q^m X_q^{(m)}$, the conditional density is $q^{-m} f_q(X_q^{(m)})$. Divide by $f_q(X_q)$ and take logarithms. Both $X_q^{(m)}$ and X_q have marginal density f_q, proving the mean-zero identity. □

Define the surprisal random variable and varentropy

$$J_q := -\log f_q(X_q), \qquad V_q := \mathrm{Var}(J_q). \tag{56}$$

Flatness of f_q at the endpoints implies that every polynomial moment of J_q is finite. Indeed the endpoint mass decays more rapidly than every algebraic scale, while $|\log f_q|$ grows at the lower-Lambert quadratic-logarithmic scale.

Theorem 6.2 (Independent-surprisal limit)**.** *Under the natural coupling in which the prefix digits and the tail copy are independent,*

$$\imath_m(X_q; S_{q,m}) - m\log(1/q) \xrightarrow[d]{m\to\infty} J_q^{(1)} - J_q^{(2)}, \tag{57}$$

where $J_q^{(1)}, J_q^{(2)}$ are independent copies of J_q. In particular,

$$\mathrm{Var}(\imath_m(X_q; S_{q,m})) \longrightarrow 2V_q. \tag{58}$$

For $|t| < 1$, the limiting cumulant generating function is

$$\boxed{\Lambda_q(t) = \log \int_{-1}^{1} f_q(x)^{1+t}\,\mathrm{d}x + \log \int_{-1}^{1} f_q(x)^{1-t}\,\mathrm{d}x.} \tag{59}$$

Proof. The prefix $S_{q,m}$ converges almost surely to a copy $\widetilde{X}_q$ built from the prefix innovations, and $\widetilde{X}_q$ is independent of the tail copy $X_q^{(m)}$. Since $q^m X_q^{(m)} \to 0$ uniformly, $X_q = S_{q,m} + q^m X_q^{(m)} \to \widetilde{X}_q$. On the open support, f_q is continuous and positive; endpoint events have probability zero. Hence the centered part of (54) converges almost surely to $\log f_q(X_q^{(m)}) - \log f_q(\widetilde{X}_q)$, a difference of independent copies. Endpoint flatness supplies uniform integrability for every fixed polynomial moment, proving (58). Independence gives

$$\mathbb{E}e^{t(J_q^{(1)} - J_q^{(2)})} = \left(\int f_q^{1-t}\right)\left(\int f_q^{1+t}\right),$$

which is finite for $|t| < 1$ and yields (59). □

Corollary 6.3 (Symmetry of all limiting information cumulants)**.** *The function Λ_q is even. Therefore every odd limiting centered cumulant vanishes, while*

$$\lim_{m\to\infty} \kappa_{2r}(\imath_m - m\log(1/q)) = 2\kappa_{2r}(J_q). \tag{60}$$

Proof. A difference of two iid random variables is symmetric. Equivalently, (59) is invariant under $t \mapsto -t$. □

Remark 6.4 (Rényi interpretation)*.* Writing $H_\alpha(f) = (1-\alpha)^{-1}\log\int f^\alpha$, formula (59) is a symmetric combination of Rényi entropies of orders $1+t$ and $1-t$. Thus the entire limiting information spectrum is encoded by the Rényi curve of the Rvachev density, not only by its Shannon entropy.

6.2 The complementary tail channel

The final variable is the sum of an independent prefix and tail. Hence

$$I(X_q; q^m X_q^{(m)}) = h(X_q) - h(S_{q,m}). \tag{61}$$

The prefix information grows linearly with m, while the information carried by the unresolved tail decays exponentially. The next section identifies its exact first asymptotic coefficient.

7 Finite-prefix entropy and Fisher information

7.1 Monotone entropy convergence

Let

$$h_{q,m} := h(S_{q,m}) = -\int b_{q,m}(x) \log b_{q,m}(x)\, \mathrm{d}x. \tag{62}$$

Proposition 7.1 (Monotone finite-sinc entropy)**.** *For every $0 < q < 1$,*

$$h_{q,1} < h_{q,2} < \cdots < h(X_q), \qquad h_{q,m} \longrightarrow h(X_q). \tag{63}$$

Moreover the entropy gap is the tail mutual information in (61).

Proof. The recursion $S_{q,m+1} = S_{q,m} + (1-q)q^m U_m$ adds a nondegenerate independent continuous summand. The entropy-power inequality therefore gives strict entropy increase. The full variable satisfies $X_q = S_{q,m} + q^m X_q^{(m)}$, so the same inequality gives $h_{q,m} < h(X_q)$. Finite convolution densities converge to f_q in L^1 and are uniformly bounded by the supremum of the first uniform density. Their supports lie in a fixed compact interval. The function $-u \log u$ is uniformly integrable on this bounded range, which yields entropy convergence. The mutual-information identity follows from independence of the prefix and tail. □

Entropy power already forces a useful lower bound. Since $X_q = S_{q,m} + q^m X_q^{(m)}$ and $h(q^m X_q^{(m)}) = h(X_q) + m \log q$,

$$h(X_q) - h_{q,m} \geq -\frac{1}{2} \log(1 - q^{2m}) = \frac{1}{2} q^{2m} + O(q^{4m}). \tag{64}$$

The exact coefficient is source-dependent and is governed by Fisher information.

7.2 A reverse-small-noise expansion

For a smooth density f, write

$$\mathcal{J}(f) = \int \frac{f'(x)^2}{f(x)}\, \mathrm{d}x. \tag{65}$$

The following technical lemma isolates the analytic step. It is phrased for the geometric-uniform family, whose compact support, endpoint flatness, and super-polynomial Fourier decay make the hypotheses automatic.

Lemma 7.2 (Fourier deconvolution at one geometric scale)**.** *As $m \to \infty$,*

$$b_{q,m} = f_q - \frac{\sigma_q^2 q^{2m}}{2} f_q'' + r_{q,m}, \tag{66}$$

where $r_{q,m} = o(q^{2m})$ in every fixed Sobolev norm and in the weighted norms needed to differentiate entropy. In particular,

$$\int r_{q,m}(x)\, \mathrm{d}x = 0. \tag{67}$$

Proof. From (41),

$$\widehat{b}_{q,m}(t) = \frac{\phi_q(t)}{\phi_q(q^m t)}.$$

Near the origin,

$$\phi_q(z) = 1 - \frac{\sigma_q^2 z^2}{2} + O(z^4), \qquad \frac{1}{\phi_q(z)} = 1 + \frac{\sigma_q^2 z^2}{2} + O(z^4).$$

Thus on $|t| \le q^{-\eta m}$ with any fixed $0 < \eta < 1$,

$$\widehat{b}_{q,m}(t) = \phi_q(t)\left(1 + \frac{\sigma_q^2 q^{2m} t^2}{2}\right) + o(q^{2m})$$

in every polynomially weighted L^1 norm. On the complementary frequencies, use the faster-than-polynomial decay of the infinite sinc product, choosing the weight and then η so that the tail is $o(q^{2m})$. Fourier inversion and $\widehat{f_q''}(t) = -t^2\phi_q(t)$ give (66). Flatness at ± 1 allows the same split after multiplication by $1 + |\log f_q|$ and controls the thin support layer where $b_{q,m} = 0$ but $f_q > 0$. Finally both densities have total mass one, giving (67). ☐

Theorem 7.3 (Fisher coefficient of the finite-sinc entropy gap)**.** *For every fixed* $0 < q < 1$,

$$\boxed{h(X_q) - h(S_{q,m}) = \frac{\sigma_q^2 \mathcal{J}(f_q)}{2}\, q^{2m} + o(q^{2m}).} \tag{68}$$

Equivalently, the unresolved-tail information satisfies

$$I(X_q; q^m X_q^{(m)}) = \frac{\sigma_q^2 \mathcal{J}(f_q)}{2}\, q^{2m} + o(q^{2m}). \tag{69}$$

Proof. Let $\Delta_m = b_{q,m} - f_q$. By Theorem 7.2,

$$\Delta_m = -\frac{\sigma_q^2 q^{2m}}{2} f_q'' + o(q^{2m}).$$

The first variation of $h(f) = -\int f \log f$ in a zero-mass direction is $-\int \Delta_m \log f_q$. The quadratic remainder is $o(q^{2m})$ in the weighted norm supplied by the lemma. Therefore

$$\begin{aligned} h(S_{q,m}) - h(X_q) &= -\int \Delta_m \log f_q + o(q^{2m}) \\ &= \frac{\sigma_q^2 q^{2m}}{2} \int f_q'' \log f_q + o(q^{2m}). \end{aligned}$$

Endpoint flatness removes the boundary term, and integration by parts gives

$$\int f_q'' \log f_q = -\int \frac{f_q'^2}{f_q} = -\mathcal{J}(f_q).$$

Rearranging proves (68); (69) follows from (61). ☐

Corollary 7.4 (Consistency with entropy power and strict non-Gaussianity)**.** *The Cramér–Rao inequality gives*

$$\sigma_q^2 \mathcal{J}(f_q) \ge 1, \tag{70}$$

so the coefficient in (68) *is at least* $1/2$*, exactly matching the leading entropy-power lower bound* (64)*. Equality would force a Gaussian score and is impossible for the compactly supported* f_q*; hence*

$$\frac{\sigma_q^2 \mathcal{J}(f_q)}{2} > \frac{1}{2}. \tag{71}$$

Proof. For a centered differentiable density, integration by parts gives $\mathbb{E}[X_q(f_q'/f_q)(X_q)] = -1$. Cauchy–Schwarz yields $1 \le \sigma_q^2 \mathcal{J}(f_q)$. Equality characterizes a linear score and hence a Gaussian density. ☐

7.3 A new Fisher bridge for F, G, and up

At $q = 1/2$, the coefficient in (68) is $\mathcal{J}(\mathrm{up})/18$. The next identity places it directly in the Fabius and inverse-Fabius coordinates.

Theorem 7.5 (Fabius–quantile representation of Rvachev Fisher information)**.** *The Fisher information of the centered Rvachev density has the equivalent forms*

$$\mathcal{J}(\mathrm{up}) = 8\int_0^1 \frac{\mathrm{up}(2x-1)^2}{\mathrm{up}(x)}\,\mathrm{d}x, \tag{72}$$

$$= \int_0^1 \frac{F'(x)^2}{F(x)(1-F(x))}\,\mathrm{d}x, \tag{73}$$

$$= \int_0^1 \frac{\mathrm{d}y}{y(1-y)G'(y)}. \tag{74}$$

All three integrals are finite.

Proof. For $0 < x < 1$, (10) and the symmetry of F' give

$$\mathrm{up}(x) = 1 - F(x), \qquad \mathrm{up}'(x) = -F'(x) = -2\,\mathrm{up}(2x-1).$$

Evenness of up therefore yields

$$\mathcal{J}(\mathrm{up}) = 2\int_0^1 \frac{F'(x)^2}{1-F(x)}\,\mathrm{d}x = 8\int_0^1 \frac{\mathrm{up}(2x-1)^2}{\mathrm{up}(x)}\,\mathrm{d}x.$$

By $x \mapsto 1-x$ and $F(1-x) = 1-F(x)$,

$$\int_0^1 \frac{F'(x)^2}{1-F(x)}\,\mathrm{d}x = \int_0^1 \frac{F'(x)^2}{F(x)}\,\mathrm{d}x.$$

Adding the two equal integrals gives (73). Finally substitute $y = F(x)$, so $x = G(y)$, $F'(x) = 1/G'(y)$, and $\mathrm{d}x = G'(y)\,\mathrm{d}y$. This gives (74). Finiteness follows from the flat endpoint asymptotics: although $1/[y(1-y)]$ diverges, G' diverges rapidly enough to compensate. □

Corollary 7.6 (Dyadic finite-sinc entropy coefficient)**.** *For* $S_m = \sum_{k=1}^{m} 2^{-k} U_{k-1}$,

$$h(\mathrm{up}) - h(S_m) = \frac{\mathcal{J}(\mathrm{up})}{18}\,4^{-m} + o(4^{-m}). \tag{75}$$

Equivalently,

$$\lim_{m\to\infty} 4^m\,[h(\mathrm{up}) - h(S_m)] = \frac{1}{18}\int_0^1 \frac{\mathrm{d}y}{y(1-y)G'(y)}. \tag{76}$$

Remark 7.7 (A scale duality)*.* The posterior Fisher information in (26) grows as q^{-2m}, whereas the information carried by the unresolved tail decays as q^{2m}. Their product has the depth-independent leading value

$$\mathcal{J}(X_q \mid S_{q,m})\, I(X_q; q^m X_q^{(m)}) \longrightarrow \frac{1}{2}\mathcal{J}(f_q)^2\sigma_q^2. \tag{77}$$

This is another exact manifestation of the reciprocal geometric scales that pervade the repository's q-analog and inverse constructions.

8 Inverse-Fabius endpoint information on the lower-Lambert scale

8.1 Imported inverse asymptotics

Let $y \downarrow 0$ and put

$$T = \log(1/y), \qquad L = \log 2, \qquad \rho = \sqrt{\frac{2T}{L}}. \tag{78}$$

The audited inverse-Fabius analysis proves the leading quantile law

$$G(y) \sim \frac{\sqrt{T}}{\mathrm{e}\sqrt{L}} \exp\bigl(-\sqrt{2LT}\bigr), \tag{79}$$

and the logarithmic elasticity expansion

$$\frac{yG'(y)}{G(y)} = \frac{1}{\rho} + O\left(\frac{1}{\rho^2}\right), \tag{80}$$

with an explicitly described period-one correction at the next order. We now transport these inverse results into local entropy and surprisal moments.

Lemma 8.1 (Endpoint quantile slope)**.** *As $y \downarrow 0$,*

$$\boxed{\log G'(y) = T - \sqrt{2LT} - 1 - \frac{L}{2} + o(1).} \tag{81}$$

The first phase-dependent term enters below the displayed constant scale.

Proof. From (80),

$$G'(y) = \frac{G(y)}{y\rho}\left(1 + O(\rho^{-1})\right).$$

Taking logarithms and using (79) gives

$$\log G'(y) = \left(\frac{1}{2}\log T - 1 - \frac{1}{2}\log L - \sqrt{2LT}\right) + T - \log\rho + o(1).$$

Since $\log\rho = \frac{1}{2}\log(2T/L)$, all $\log T$ and $\log L$ terms cancel except $-\frac{1}{2}\log 2 = -L/2$, proving (81). $\square$

8.2 Entropy contained in an endpoint probability tail

The lower endpoint of the Fabius density F' contains probability y on the interval $[0, G(y)]$. Define its entropy contribution by

$$\mathscr{H}_F(y) := -\int_0^{G(y)} F'(x)\log F'(x)\,\mathrm{d}x. \tag{82}$$

Substitution $u = F(x)$ gives the exact quantile identity

$$\mathscr{H}_F(y) = \int_0^y \log G'(u)\,\mathrm{d}u. \tag{83}$$

Theorem 8.2 (Lower-Lambert tail entropy)**.** *As $y \downarrow 0$,*

$$\boxed{\mathscr{H}_F(y) = y\left(T - \sqrt{2LT} - \frac{L}{2} + o(1)\right).} \tag{84}$$

For the centered Rvachev density, let $\mathscr{H}_{\mathrm{up}}(y)$ denote the entropy contribution of either endpoint interval carrying probability y. Then

$$\boxed{\mathscr{H}_{\mathrm{up}}(y) = y\left(T - \sqrt{2LT} + \frac{L}{2} + o(1)\right).} \tag{85}$$

Proof. Write $u = \mathrm{e}^{-t}$. Equation (83) becomes

$$\mathscr{H}_F(y) = \int_T^\infty \mathrm{e}^{-t}\left(t - \sqrt{2Lt} - 1 - \frac{L}{2} + o(1)\right) \mathrm{d}t.$$

Watson integration at the lower endpoint gives

$$\int_T^\infty \mathrm{e}^{-t} t \, \mathrm{d}t = y(T+1), \qquad \int_T^\infty \mathrm{e}^{-t}\sqrt{2Lt}\, \mathrm{d}t = y\left(\sqrt{2LT} + o(1)\right).$$

The $+1$ from the first integral cancels the explicit -1 in (81), proving (84). If $X \sim \mathrm{up}$ and $Y = (X+1)/2$, then F' is the density of Y and $\mathrm{up}(x) = \frac{1}{2}F'((x+1)/2)$. Therefore the centered surprisal exceeds the Fabius-density surprisal by L, adding Ly to the tail contribution and proving (85). ☐

Corollary 8.3 (Endpoint surprisal moments)**.** *For every fixed positive integer r,*

$$\int_0^y \left[\log G'(u)\right]^r \mathrm{d}u = yT^r\left(1 - r\sqrt{\frac{2L}{T}} + O\left(\frac{1}{T}\right)\right). \tag{86}$$

In particular, the unnormalized second surprisal moment is

$$\int_0^y \left[\log G'(u)\right]^2 \mathrm{d}u = y\left(T^2 - 2\sqrt{2L}\, T^{3/2} + O(T)\right). \tag{87}$$

Proof. By (81), $\log G'(\mathrm{e}^{-t}) = t - \sqrt{2Lt} + O(1)$. Raise this to the fixed power r, retain the first two orders, and apply the same Watson integration. Differentiation of t^r contributes only relative order $1/T$. ☐

Remark 8.4 (Where the periodic Lambert phase should reappear)*.* The displayed entropy law averages over a probability interval and therefore pushes the phase ripple below the constant term. The audited elasticity has an explicit periodic correction of order ρ^{-2}. Carrying Theorem 8.1 two orders further should produce a periodic contribution to $\mathscr{H}_F(y)/y$ at order $T^{-1/2}$. This is formulated precisely as Theorem 11.3 below.

9 The near-Gaussian $q \uparrow 1$ frontier

9.1 Exact standardized Bernoulli cumulants

Set

$$Z_q = \frac{X_q}{\sigma_q}, \qquad \mathrm{Var}(Z_q) = 1, \tag{88}$$

and let $\lambda_r(q) = \kappa_r(Z_q)$ be its standardized cumulants. Combining (6) and (5) gives an exact rational q-formula.

Theorem 9.1 (Standardized cumulant hierarchy)**.** *For $r \geq 2$,*

$$\boxed{\lambda_{2r}(q) = \frac{3^r 2^{2r} B_{2r}}{2r} \frac{(1-q^2)^r}{1-q^{2r}} = \frac{3^r 2^{2r} B_{2r}}{2r} \frac{(1-q^2)^{r-1}}{1+q^2+\cdots+q^{2r-2}}.} \tag{89}$$

All odd standardized cumulants vanish. In particular,

$$\lambda_4(q) = -\frac{6}{5}\frac{1-q^2}{1+q^2}, \tag{90}$$

$$\lambda_6(q) = \frac{48}{7}\frac{(1-q^2)^2}{1+q^2+q^4}. \tag{91}$$

Thus $\lambda_{2r}(q) = O((1-q)^{r-1})$ as $q \uparrow 1$.

Proof. Divide (6) by σ_q^{2r} and simplify using $1 - q^{2r} = (1-q^2)(1+q^2+\cdots+q^{2r-2})$. □

The exact hierarchy explains why the Gaussian limit is unusually orderly: at each new order in $1-q$, only finitely many Bernoulli cumulants contribute.

9.2 Formal Hermite calculation of the entropy deficit

Let $\gamma(x) = (2\pi)^{-1/2}\mathrm{e}^{-x^2/2}$ and

$$\Delta(q) = D(Z_q\|\gamma) = \frac{1}{2}\log(2\pi\mathrm{e}) - h(Z_q) = \delta_q. \tag{92}$$

Use the probabilists' Hermite polynomials H_n. A formal Edgeworth density ratio through the orders relevant for cubic entropy is

$$\frac{f_{Z_q}(x)}{\gamma(x)} = 1 + \frac{\lambda_4}{4!}H_4(x) + \frac{\lambda_6}{6!}H_6(x) + \frac{\lambda_4^2}{2(4!)^2}H_8(x) + O((1-q)^3). \tag{93}$$

Hermite orthogonality eliminates the cross terms involving H_6 and H_8 at cubic order. Since

$$\mathbb{E}H_4(N)^2 = 4! = 24, \qquad \mathbb{E}H_4(N)^3 = 1728, \qquad N \sim \mathcal{N}(0,1), \tag{94}$$

the formal entropy expansion is

$$\Delta(q) = \frac{\lambda_4(q)^2}{48} - \frac{\lambda_4(q)^3}{48} + O((1-q)^4). \tag{95}$$

Writing $\epsilon = 1-q$ and expanding (90) gives

$$\boxed{\Delta(q) = \frac{3}{100}\epsilon^2 + \frac{33}{500}\epsilon^3 + O(\epsilon^4).} \tag{96}$$

The exact symbolic derivation is reproduced by `data/symbolic_edgeworth.txt` in the archive.

Conjecture 9.2 (Entropic Edgeworth theorem for geometric uniforms)**.** *Equation (96) is a genuine asymptotic expansion as $q \uparrow 1$. More generally,*

$$\Delta(q) \sim \sum_{n \geq 2} d_n (1-q)^n, \tag{97}$$

where every d_n is an explicitly computable rational polynomial in Bernoulli numbers. Truncation after N terms has a uniform remainder $O_N((1-q)^{N+1})$.

Remark 9.3 (Why this remains labeled conjectural). The product formula and Theorem 9.1 make the formal algebra finite at each order. What remains is not coefficient discovery but uniform analysis: one must control the Edgeworth approximation in the moderate and far tails strongly enough to integrate $f \log(f/\gamma)$. General entropic Edgeworth theorems strongly support this conclusion, but the geometric triangular array and its exact compact support deserve a dedicated proof rather than a silent appeal to an iid theorem.

Conjecture 9.4 (Monotonic Gaussianization)**.** *The entropy deficit δ_q is strictly decreasing on* $(0, 1)$*, with*

$$\lim_{q \downarrow 0} \delta_q = \frac{1}{2} \log \frac{\pi \mathrm{e}}{6}, \qquad \lim_{q \uparrow 1} \delta_q = 0. \tag{98}$$

Equivalently, after variance normalization, adding more nearly equal geometric scales increases entropy monotonically toward the Gaussian optimum.

The endpoint limits are proved: the first follows from convergence to a uniform law and the second from the triangular-array central limit theorem plus entropic regularity. Strict monotonicity between them is the open part. A promising route is to compare coefficient vectors by majorization and seek a strict entropy version of the convex-order monotonicity already developed in the repository.

10 Numerical experiments

10.1 Method and reproducibility

The accompanying script `experiments.py` does not call a black-box Fabius function. It iterates the exact CDF transfer operator obtained from (4) on a uniform grid of 200001 points. If F is the CDF of a trial density, one iteration is

$$(\mathcal{T}_q f)(x) = \frac{F((x + 1 - q)/q) - F((x - 1 + q)/q)}{2(1 - q)}. \tag{99}$$

The moving-CDF form is linear-time in the grid size and remains stable for q close to one. Finite prefix densities are computed by the analogous sequence of moving averages, avoiding catastrophic cancellation in the exact Thue–Morse formula (47). Fisher information uses a centered finite-difference score with a documented endpoint floor; the omitted mass is far below the grid error because the density is flat. Symbolic cumulant and Hermite calculations use exact SymPy rationals. Monte Carlo experiments use the fixed seed `20260830`.

The principal command is

```
python experiments.py --output . --grid-points 200001 --sample-size 180000
```

The generated CSV files contain more digits than displayed below.

10.2 Entropy deficit and the $q \uparrow 1$ coefficients

Table 2: Numerical information functionals for selected geometric ratios. The last ratio should tend to $3/100 = 0.03$ if Theorem 9.2 holds.

q	$h(X_q)$	δ_q	$\delta_q/(1-q)^2$	$\mathcal{J}(f_q)$	V_q
0.50	0.297039913	2.3286332×10^{-2}	0.0931453	11.8007	0.259695
0.70	-0.003097635	5.4294958×10^{-3}	0.0603277	17.9710	0.374413
0.80	-0.230888773	1.9088735×10^{-3}	0.0477218	27.4875	0.424614
0.90	-0.602963697	3.7659604×10^{-4}	0.0376596	57.1846	0.466401
0.95	-0.962232301	8.3866853×10^{-5}	0.0335467	117.0811	0.484137
0.97	-1.222692142	2.8810502×10^{-5}	0.0320117	197.0463	0.490695

For $q = 0.95$, the cubic prediction in (96) is 8.3250×10^{-5}; the computed value is 8.3867×10^{-5}. For $q = 0.97$, the prediction is 2.8782×10^{-5} and the computed value is 2.8811×10^{-5}.

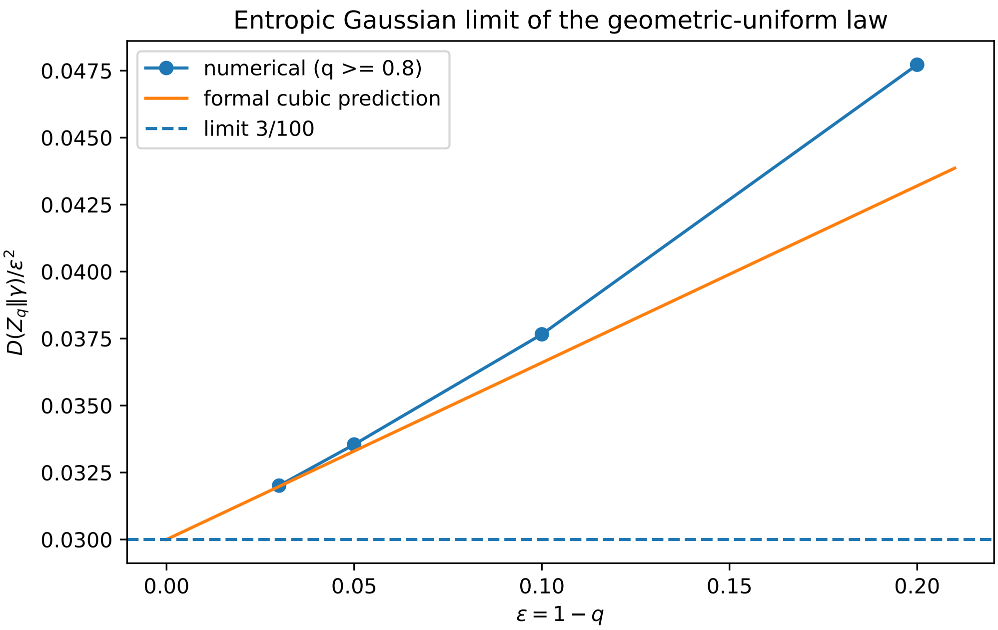

Figure 1: The scaled entropy deficit approaches the predicted limit 3/100. The orange line is the formal cubic correction. Coarser q values remain in the CSV but are omitted from the asymptotic comparison.

10.3 Dyadic entropy, Fisher information, and finite prefixes

For the centered Rvachev density the computation gives

$$h(\mathrm{up}) = 0.297039912858918, \tag{100}$$
$$\mathcal{J}(\mathrm{up}) = 11.800710562320129, \tag{101}$$
$$V(\mathrm{up}) = 0.259694517369981, \tag{102}$$
$$\delta_{1/2} = 0.023286331677645 \text{ nats} = 0.033595075231834 \text{ bits}. \tag{103}$$

Thus the predicted finite-prefix coefficient is

$$\frac{\mathcal{J}(\mathrm{up})}{18} = 0.655595031240007. \tag{104}$$

Table 3: Entropy convergence of the dyadic finite sinc prefixes.

m	$h(S_m)$	$h(\mathrm{up}) - h(S_m)$	$4^m[h(\mathrm{up}) - h(S_m)]$	exact MMSE
1	0.000000000	2.9703991×10^{-1}	1.1881597	2.7777778×10^{-2}
2	0.250000000	4.7039913×10^{-2}	0.7526386	6.9444444×10^{-3}
3	0.286463907	1.0576006×10^{-2}	0.6768644	1.7361111×10^{-3}
4	0.294459271	2.5806416×10^{-3}	0.6606443	4.3402778×10^{-4}
5	0.296398474	6.4143912×10^{-4}	0.6568337	1.0850694×10^{-4}
6	0.296879781	1.6013200×10^{-4}	0.6559007	2.7126736×10^{-5}
7	0.296999895	4.0018269×10^{-5}	0.6556593	6.7816840×10^{-6}
8	0.297029910	1.0002877×10^{-5}	0.6555485	1.6954210×10^{-6}

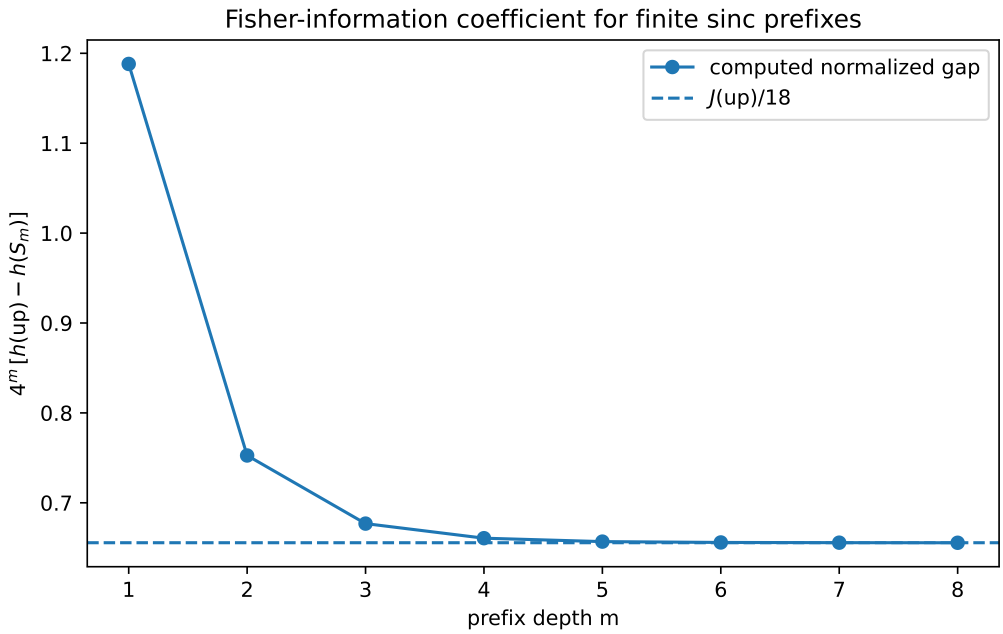

Figure 2: The normalized finite-prefix entropy gap converges rapidly to the Fisher coefficient $\mathcal{J}(\mathrm{up})/18$. Depths beyond eight are retained in the CSV, but their gaps approach the fixed-grid quadrature resolution.

10.4 Information-spectrum convergence

The limiting variance predicted by (58) is

$$2V(\mathrm{up}) = 0.519389034739962. \tag{105}$$

Table 4: Monte Carlo information-spectrum check with 180000 samples. The KS column compares the centered information density with a separately simulated difference of independent Rvachev surprisals.

m	centered mean	centered variance	two-sample KS distance
2	-0.001960	0.472869	0.004522
4	0.001938	0.521161	0.003367
8	-0.001691	0.521082	0.004867

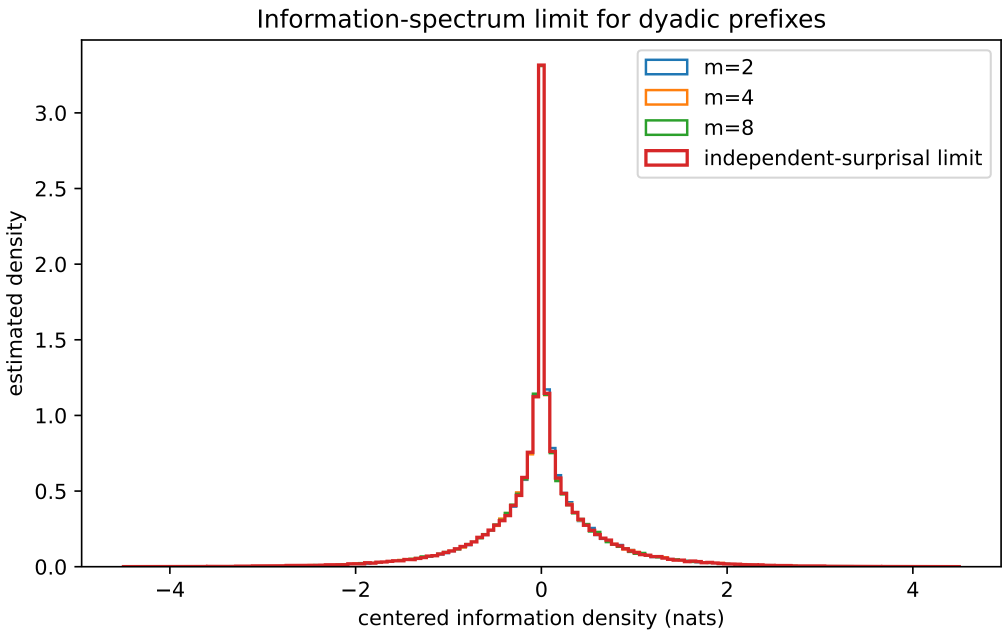

Figure 3: Centered information densities at depths $2, 4, 8$ and the independent-surprisal limit. The pronounced central cusp is a genuine non-Gaussian feature; convergence is not toward a normal information spectrum.

10.5 Numerical interpretation

The experiments support four distinct statements, only one of which is conjectural:

1. The exact mutual-information and MMSE laws are symbolic identities; the numerics merely provide regression tests.
2. The Fisher coefficient in (75) is strongly confirmed by the stabilized normalized gaps in Table 3.
3. The distributional limit in Theorem 6.2 is confirmed by both variance and KS diagnostics.
4. The entropic coefficients in (96) agree with the $q = 0.95$ and $q = 0.97$ computations, but the theorem-level uniform remainder remains open as stated in Theorem 9.2.

11 Conjectures and a prioritized frontier program

11.1 Full information-spectrum asymptotics

Conjecture 11.1 (Quantitative surprisal convergence). *For every compact interval $K \subset (-1, 1)$ in the transform variable t,*

$$\sup_{t \in K} \left| \log \mathbb{E} e^{t(\imath_m - R_m)} - \Lambda_q(t) \right| = O_q(q^m), \tag{106}$$

with an all-orders expansion whose coefficients are integrals of score polynomials in f_q. The same rate should hold in Wasserstein distance for the centered information density.

A direct route is to write $X_q = \widetilde{X}_q + q^m (X_q^{(m)} - \widetilde{X}_q^{(m)})$ under an independent-copy coupling and

Taylor-expand $\log f_q$ away from the endpoints. The lower-Lambert endpoint estimates should make the omitted boundary probability smaller than every power of q^m.

11.2 Optimal coding versus the exact geometric prefix

Problem 11.2 (Exact rate distortion at self-similar points)**.** *Determine whether $R_{X_q}(D_m)$ admits an expansion*

$$R_{X_q}(D_m) = R_m - \delta_q + c_1(q)q^{\alpha_1 m} + c_2(q)q^{\alpha_2 m} + \cdots , \tag{107}$$

with explicit exponents and coefficients. Decide whether an optimal reproduction can be chosen from a corrected geometric prefix $S_{q,m} + q^m \Psi_q(S_{q,m})$.

The exact posterior makes this unusually concrete: the test channel is known at all resolutions, and the leading suboptimality is exactly δ_q. A high-resolution perturbation of the posterior centroid or companding map may expose the next coefficient.

11.3 Periodic Lambert corrections to tail information

Conjecture 11.3 (Phase-resolved endpoint entropy)**.** *Let Ψ be the period-one lower-Lambert phase from the repository's all-orders inverse expansion. There exists an explicit period-one function Θ such that*

$$\frac{\mathscr{H}_F(y)}{y} = T - \sqrt{2LT} - \frac{L}{2} + \frac{\Theta(\rho+\beta)}{\sqrt{T}} + O\left(\frac{\log T}{T}\right), \tag{108}$$

where $\rho = \sqrt{2T/L}$ and $\beta = 1/L + 1/2$. The Fourier coefficients of Θ should be explicit Gamma–zeta multiples inherited from Ψ and its derivative.

This is a clean test of how much log-periodicity survives one entropy integration. The calculation requires no new saddle point: insert the audited all-orders elasticity into (83) and apply a derivative-stable Watson expansion.

11.4 A Bayesian spectral operator

Define the normalized posterior pullback

$$(\mathcal{B}_m g)(s) := \mathbb{E}\left[g\left(\frac{X_q - S_{q,m}}{q^m}\right) \middle| S_{q,m} = s \right]. \tag{109}$$

By self-similarity, $\mathcal{B}_m g$ is the constant $\int g f_q$, so the normalized residual erases the prefix. More interesting is the unnormalized conditional expectation

$$(\mathcal{C}_m g)(s) = \mathbb{E}[g(X_q) \mid S_{q,m} = s] = \int_{-1}^{1} g(s + q^m z) f_q(z)\, \mathrm{d}z. \tag{110}$$

Problem 11.4 (q-Appell/Legendre singular system)**.** *Diagonalize or triangularize $\mathcal{C}_m$ simultaneously in the repository's Appell and Legendre bases. On polynomials, translation and the exact moments make $\mathcal{C}_m$ finite triangular. Determine whether its finite determinants, singular values, or exterior powers factor into q-Pochhammer products and Gaussian-binomial coefficients.*

The leading diagonal in an Appell basis is expected to contain q^{mn}, while the translation part should generate a q-Weyl algebra closely related to the existing Koopman calculus. This would connect the present Bayesian filtration to the repository's transfer determinants without duplicating the earlier operator results.

11.5 Thue–Morse information identities beyond KL

Problem 11.5 (Finite Thue–Morse Rényi information). *Evaluate or simplify the order-α Rényi mutual information of the dyadic prefix channel:*

$$I_\alpha(S_m; X) = \frac{1}{\alpha - 1} \log \iint b_m(s) \left(\frac{2^m \operatorname{up}(2^m(x-s))}{\operatorname{up}(x)} \right)^\alpha \operatorname{up}(x) \, \mathrm{d}x \, \mathrm{d}s. \tag{111}$$

At $\alpha = 1$ it collapses to $m \log 2$. At integer orders, substitution of the finite Thue–Morse spline may produce exact multi-sums controlled by Prouhet cancellation and the Rvachev moment algebra.

A particularly tractable case is the collision information $\alpha = 2$, where Parseval and the finite/infinite sinc products may turn the double integral into a single Fourier integral with valuation-controlled zeros.

11.6 General masks and higher dimensions

Conjecture 11.6 (Equal-information affine masks). *Let a compactly supported law satisfy an independent affine refinement*

$$X \overset{d}{=} AX' + \xi, \tag{112}$$

with invertible contraction $A \in \mathbb{R}^{d \times d}$ and an absolutely continuous innovation ξ. For the depth-m innovation prefix S_m,

$$I(X; S_m) = -m \log |\det A|, \tag{113}$$

provided the relevant differential entropies are finite. The posterior covariance is $A^m \Sigma (A^m)^\top$, and the finite-prefix entropy gap has leading term

$$\frac{1}{2} \operatorname{tr} \left(A^m \Sigma (A^m)^\top \mathcal{J}_f \right), \tag{114}$$

where $\mathcal{J}_f$ is the Fisher information matrix.

The mutual-information part follows formally by the same entropy-scaling proof; the frontier is to formulate minimal hypotheses and connect matrix masks to multivariate Rvachev atomic functions, box splines, Hadamard masks, and multivariate Thue–Morse systems.

11.7 Formalization priorities

The exact finite layer is well suited to Lean. A practical order is:

1. define the geometric prefix and prove the independent tail decomposition;
2. formalize entropy under translation and nonzero scalar multiplication;
3. prove Theorems 3.1, 3.2 and 4.1;
4. specialize to $q = 1/2$ and connect the finite prefix to the existing Thue–Morse spline and Rvachev density APIs;
5. formalize the posterior cumulant and Bell-moment scaling;
6. prove the Fisher bridge Theorem 7.5 using one-dimensional change of variables;
7. postpone Theorem 7.3 until a reusable entropy first-variation and Fourier-tail library is available;

8. treat the lower-Lambert entropy theorem only after importing the existing phase-aware inverse asymptotic interface.

The first five steps are finite or elementary measure-theoretic identities and can serve as robust regression tests for later analytic formalization.

12 Conclusion

The geometric random series behind the Fabius and Rvachev functions carries an information structure as rigid as its arithmetic and harmonic structures. A finite prefix is simultaneously a finite sinc product, a box spline, a q-cumulant truncation, and a sufficient statistic. The self-similar tail makes the information filtration exactly linear:

$$I(X_q; S_{q,m}) = m \log(1/q).$$

The same identity yields a compact non-Gaussian channel obeying the Gaussian correlation formula, exact MMSE, and a self-similar rate-distortion curve. At $q = 1/2$, the channel density is an explicit Thue–Morse spline and the likelihood is a shifted Rvachev atom, closing a finite/infinite sinc loop through an exact KL integral.

The fluctuation around the linear information law is not Gaussian. It converges to a difference of independent surprisals, so varentropy and the complete Rényi curve become natural new invariants of the Fabius–Rvachev law. The unresolved tail carries exponentially small information, whose first coefficient is Fisher information. This creates a new transform bridge

$$\mathcal{J}(\mathrm{up}) = \int_0^1 \frac{\mathrm{d}y}{y(1-y)G'(y)}$$

and a new asymptotic bridge from finite sinc prefixes to the inverse Fabius function. At the flat endpoint, the lower-Lambert quantile scale governs not only values and derivatives but the entropy contained in a probability tail.

Finally, the exact Bernoulli cumulants organize the near-Gaussian regime. The first entropic coefficients $3/100$ and $33/500$ emerge by a finite Hermite calculation and are accurately reproduced numerically. Proving the full uniform entropic expansion, resolving the periodic tail-entropy correction, and constructing the Bayesian Appell–Legendre spectral system are concrete next steps. Together they extend the repository's existing arithmetic, inverse, q-series, sinc, and representation theories in a genuinely different direction: from exact values and transforms to exact information geometry.

A Detailed derivations and auxiliary identities

A.1 Entropy-power calculation for the deficit bound

For completeness, if $a_j = (1-q)q^j$ and $U_j \sim \mathrm{Unif}[-1,1]$, then

$$N(a_j U_j) = a_j^2 \frac{\mathrm{e}^{2\log 2}}{2\pi\mathrm{e}} = a_j^2 \frac{2}{\pi\mathrm{e}}.$$

The entropy-power inequality gives

$$N(X_q) \geq \frac{2}{\pi\mathrm{e}} \sum_{j\geq 0} a_j^2 = \frac{2(1-q)}{\pi\mathrm{e}(1+q)}.$$

Thus

$$h(X_q) \geq \frac{1}{2}\log\left(2\pi\mathrm{e}\,\frac{2(1-q)}{\pi\mathrm{e}(1+q)}\right) = \log 2 + \frac{1}{2}\log\frac{1-q}{1+q}.$$

Subtracting from the matching Gaussian entropy

$$\frac{1}{2}\log\left(2\pi\mathrm{e}\frac{1-q}{3(1+q)}\right)$$

gives $\delta_q \leq \frac{1}{2}\log(\pi\mathrm{e}/6)$.

A.2 Exact information and correlation in one line

The prefix variance and tail variance split orthogonally:

$$\mathrm{Var}(S_{q,m}) = \sigma_q^2(1-q^{2m}), \qquad \mathrm{Var}(q^m X_q^{(m)}) = \sigma_q^2 q^{2m}.$$

Therefore

$$1 - \mathrm{Corr}(X_q, S_{q,m})^2 = \frac{\mathrm{Var}(X_q - S_{q,m})}{\mathrm{Var}(X_q)} = q^{2m},$$

while entropy scaling gives

$$I(X_q; S_{q,m}) = h(X_q) - h(q^m X_q^{(m)}) = -m\log q.$$

The two identities coincide without any optimization or Gaussian assumption.

A.3 Formal Hermite entropy algebra

Let $r = 1 + u = f_{Z_q}/\gamma$. Since $\int \gamma u = 0$,

$$D(f_{Z_q}\|\gamma) = \int \gamma(1+u)\log(1+u) = \frac{1}{2}\mathbb{E}_\gamma u^2 - \frac{1}{6}\mathbb{E}_\gamma u^3 + O(\mathbb{E}|u|^4).$$

To cubic order in $\epsilon = 1-q$, only $u_1 = (\lambda_4/24)H_4$ contributes after Hermite orthogonality. Hence

$$\frac{1}{2}\mathbb{E}u_1^2 = \frac{1}{2}\frac{\lambda_4^2}{24^2}\,24 = \frac{\lambda_4^2}{48},$$

and

$$-\frac{1}{6}\mathbb{E}u_1^3 = -\frac{1}{6}\frac{\lambda_4^3}{24^3}\,1728 = -\frac{\lambda_4^3}{48}.$$

SymPy verifies

$$\lambda_4 = -\frac{6}{5}\epsilon - \frac{3}{5}\epsilon^2 + O(\epsilon^4),$$

which gives

$$\frac{\lambda_4^2 - \lambda_4^3}{48} = \frac{3}{100}\epsilon^2 + \frac{33}{500}\epsilon^3 + O(\epsilon^4).$$

B Corpus-relative nonduplication notes

The audit compared the present claims with the following major source families in the recursive tree:

Source family	Material treated as existing input
Primary Fabius/Rvachev exposition and glossary	definitions, functional equations, exact arithmetic, probability, Fourier product, moments, corrected asymptotics
Lean walkthrough and formalization modules	exact dyadic recurrences, positivity, flatness, logarithmic delay equations, formalized asymptotic interfaces
Inverse-and-sampling drafts	inverse derivatives, Lambert W, all-orders periodic phase, quantile transforms, sampling and interpolation
Exponents and q-series drafts	q-Pochhammer, Gaussian binomial, geometric comb interpolation, inverse q-analogs, parameter asymptotics
Integral/fractional-calculus drafts	Mellin, Stieltjes, Cauchy, fractional, primitive, and global quantile-entropy identities
Representation drafts	Legendre/Jacobi/Chebyshev/Lagrange expansions, finite synthesis, multiresolution, sampling, Fredholm and Brownian representations
Atomic functions beyond dyadic	general-base convolutions, Cantor/nonanalytic sets, derivative energies, Rényi entropy, complex dimensions
Stein–Koopman report	Appell eigenmodes, q-Weyl calculus, transfer determinants, Poisson equations, martingales, Stein kernels
Recent frontier reports	negative-q duality, nonanalyticity of iterates, arithmetic-comb algorithms, additional endpoint and transform identities

No exact treatment of the geometric-prefix mutual information, posterior MMSE, information density, varentropy limit, rate-distortion prefix curve, finite-sinc entropy correction, or Fisher bridge (74) was located. Existing entropy statements concern the full quantile integral or different derivative- energy measures. The standalone file `CORPUS_AUDIT.md` records the search terms and contribution boundary in a compact form.

C Reproducibility manifest

The ZIP archive contains:

Path	Purpose
`fabius_information_frontier.tex`	complete LaTeX report
`fabius_information_frontier.pdf`	compiled PDF
`experiments.py`	commented deterministic and Monte Carlo experiment driver
`requirements.txt`	Python dependency list
`README.md`	reproduction instructions and file map
`CORPUS_AUDIT.md`	nonduplication audit summary
`data/q_entropy_table.csv`	q-dependent entropy, Fisher information, and varentropy
`data/dyadic_prefix_entropy.csv`	finite-prefix entropy and Fisher coefficient
`data/information_spectrum_summary.csv`	Monte Carlo information-spectrum diagnostics
`data/thue_morse_signs.csv`	exact finite Thue–Morse signs through depth eight
`data/symbolic_edgeworth.txt`	exact symbolic cumulant/Hermite check

Path	Purpose
`figures/*.pdf,figures/*.png`	vector and raster numerical figures
`numerical_summary.txt`	human-readable numerical constants
`SHA256SUMS`	integrity hashes for the archive contents

The script requires only NumPy, SymPy, and Matplotlib. No network access is required. The exact theorems do not depend on the numerical experiments.

References

[1] B. Jessen and A. Wintner, Distribution functions and the Riemann zeta function, *Transactions of the American Mathematical Society* **38** (1935), 48–88.

[2] J. Fabius, A probabilistic example of a nowhere analytic C^∞-function, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174.

[3] V. A. Rvachev, Compactly supported solutions of functional-differential equations and their applications, *Russian Mathematical Surveys* **45** (1990), no. 1, 87–120.

[4] J. Arias de Reyna, An infinitely differentiable function with compact support: definition and properties, English translation of the 1982 article, arXiv:1702.05442.

[5] J. Arias de Reyna, Arithmetic of the Fabius function, *INTEGERS* **18** (2018), Article A51; arXiv:1702.06487.

[6] J. K. Haugland, Evaluating the Fabius function, arXiv:1609.07999.

[7] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.

[8] G. Gasper and M. Rahman, *Basic Hypergeometric Series*, 2nd ed., Cambridge University Press, 2004.

[9] C. E. Shannon, A mathematical theory of communication, *Bell System Technical Journal* **27** (1948), 379–423, 623–656.

[10] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006.

[11] A. Dembo, T. M. Cover, and J. A. Thomas, Information theoretic inequalities, *IEEE Transactions on Information Theory* **37** (1991), 1501–1518.

[12] A. J. Stam, Some inequalities satisfied by the quantities of information of Fisher and Shannon, *Information and Control* **2** (1959), 101–112.

[13] A. R. Barron, Entropy and the central limit theorem, *Annals of Probability* **14** (1986), 336–342.

[14] S. G. Bobkov, G. P. Chistyakov, and F. Götze, Rate of convergence and Edgeworth-type expansion in the entropic central limit theorem, *Annals of Probability* **41** (2013), no. 4, 2479–2512.

[15] O. Johnson, *Information Theory and the Central Limit Theorem*, Imperial College Press, 2004.

[16] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, On the Lambert W function, *Advances in Computational Mathematics* **5** (1996), 329–359.

[17] V. Reshetnikov, *ProveIt: FabiusFunction documentation and research-frontier corpus*, recursive LaTeX tree under `Analysis/FabiusFunction/docs`, GitHub repository, accessed 30 August 2026, https://github.com/VladimirReshetnikov/ProveIt.